

Article

Spatiotemporally Coordinated Operation in Multiple Data Centers Based on Adaptive Large Neighborhood Search Algorithm with Hierarchical Collaboration

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Abstract

Data centers have become essential infrastructure for digital services, while their rapidly growing electricity demand makes coordinated workload and power management an important optimization problem. This paper studies the multi-data-center operation problem under time-of-use electricity pricing and formulates it as a multi-data-center mixed-integer nonlinear programming model (MDC-MINLP). The model jointly represents binary task scheduling decisions, including temporal workload shifting and spatial task migration, and continuous power-side variables, including device-level utilization, IT and auxiliary power consumption, energy storage dynamics, grid power procurement, and quality-of-service constraints. The objective is to minimize the total operating cost by integrating electricity purchasing cost, IT operation loss, storage degradation cost, and migration cost. To solve the resulting large-scale discrete–continuous coupled problem, an Adaptive Large Neighborhood Search algorithm with Hierarchical Collaboration (HC-ALNS) is proposed. HC-ALNS reconstructs feasible task action sets, employs a surrogate objective for fast candidate screening, performs accurate power-layer evaluation for selected solutions, and adaptively adjusts search intensity according to convergence behavior. Numerical results show that HC-ALNS reduces the total operating cost by 3.67% and achieves better convergence and solution quality than NSGA-II and PSO. These findings demonstrate that the proposed MDC-MINLP and HC-ALNS provide an effective mathematical optimization framework for coordinated computation–power scheduling.

Keywords: mixed-integer nonlinear programming; adaptive large neighborhood search; hierarchical optimization; discrete–continuous optimization; multi-data-center scheduling; computation–power coordination



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1. Introduction

With the rapid development of cloud computing, artificial intelligence, and big data technologies, data centers—which are critical to digital economic infrastructure—have experienced continuous growth in both scale and computing demand, leading to increasingly prominent energy consumption issues. Studies indicate that the global electricity consumption of data centers has reached approximately 415 TWh, accounting for about 1.5% of

global electricity demand, with an average annual growth rate of around 12% in recent years [1]. Due to the increasing prevalence of computation-intensive applications such as artificial intelligence, this figure is projected to increase to approximately 945 TWh by 2030, accounting for nearly 3% of global electricity demand [2,3]. Meanwhile, as data centers continue to expand in scale, their high-power-density operation further increases instantaneous power demand, not only intensifying pressure on power systems but also raising the complexity of operational decision-making [4]. Therefore, improving data center energy efficiency is not only an engineering issue but also an important mathematical optimization problem involving workload scheduling, resource allocation, and power-side coordination.

Extensive studies have been conducted on computing–power coordinated scheduling for data centers. Unlike conventional high-energy-consuming loads, data centers can achieve rapid power load shifting and regulation through the temporal and spatial migration of computing workloads, thereby providing flexible decision variables for mathematical scheduling models. The authors of [5] classify delay-tolerant workloads as short-delay, long-duration continuous, and interruptible types, and establish corresponding temporal shifting models, which are embedded into day-ahead optimal scheduling of data center microgrids to reduce energy consumption and curtailment costs. Meanwhile, in [6], the authors propose a spatiotemporal load-shifting framework based on virtual links, transforming data center load flexibility into dispatchable resources that can participate in electricity market clearing while explicitly defining pricing and compensation mechanisms. Local flexibility market mechanisms provide further support for the market-oriented utilization of load flexibility. Reference [7] provides a systematic review of the modeling and clearing methods of local flexibility markets and establishes a multi-agent coordinated trading framework. The authors of [8] develop a multi-data-center energy consumption model that considers spatiotemporal flexibility; jointly characterizes delay-tolerant task scheduling, IT equipment operation, and cooling processes; and defines an energy flexibility region to describe the feasible power consumption space. Furthermore, in [9], the authors transform the internal operational constraints of data centers into privacy-preserving flexibility region constraints, enabling the reliability of power systems to be assessed without overlooking load flexibility. These studies show that the flexible operation of data centers can be mathematically represented through schedulable regions, coupling constraints, and optimization models. However, most existing studies still focus on macroscopic load flexibility or regional flexibility representations, while the fine-grained coupling between task-level scheduling decisions and power-side operational variables remains insufficiently explored.

Scheduling problems under time-varying electricity prices have increasingly been studied from the perspective of mathematical optimization. Feng et al. [10] formulate an energy-efficient batch scheduling problem under time-of-use electricity prices as a mixed-integer linear programming model and solve it using heuristic algorithms. Zhao and Zhu [11] develop a chance-constrained stochastic mixed-integer linear programming model and a reinforcement-learning-based adaptive large neighborhood search algorithm for continuous-time energy-aware scheduling. These studies show that mathematical programming and improved heuristics are effective tools for solving large-scale scheduling problems with coupled decisions.

Although refined mathematical models can accurately capture the coupling characteristics of computing–power coordination, they usually introduce a large number of integer variables, nonlinear relationships, and strongly coupled constraints, resulting in high computational complexity and low solution efficiency. Therefore, many researchers have proposed metaheuristic and hybrid optimization methods to improve scheduling efficiency. Attiya et al. [12] propose an intelligent chimp optimizer combined with the marine

predators algorithm and a disruption operator for IoT task scheduling in fog computing, showing the potential of hybrid metaheuristics for large-scale task allocation problems. The authors of [13] propose a hybrid algorithm that combines the genetic algorithm (GA) with particle swarm optimization (PSO), where GA enhances population diversity through crossover and mutation, mitigating the premature convergence of PSO and improving both global search capability and convergence speed. In [14], the researchers develop a discrete non-dominated sorting genetic algorithm (DNSGA-II) for fog–cloud collaborative multi-objective scheduling by redesigning crossover and mutation operators to adapt to task assignment decisions. The authors of [15] propose an improved PSO combined with simulated annealing (IPSO-SA), incorporating Tent chaotic mapping for initialization and nonlinear adaptive inertia weights, along with the Metropolis criterion to avoid premature convergence. The study in [16] designs a load–energy–time-aware ant colony optimization (LET-ACO) algorithm for dynamic energy management in cloud data centers, integrating dynamic voltage scaling and load-balancing mechanisms to reduce energy consumption without affecting system performance. Other optimization algorithms, such as artificial bee colony [17], genetic optimization [18], cuckoo search [19], and chaotic cat swarm optimization [20], have also been widely studied. These methods provide important algorithmic foundations for solving large-scale scheduling problems, but most of them are general-purpose heuristics and do not fully exploit the hierarchical coupling structure between task-level scheduling and power-side operation. Therefore, a problem-oriented algorithmic framework is still needed for large-scale data center scheduling problems with discrete–continuous coupling.

Although significant progress has been achieved in computing–power coordinated optimization, several limitations remain:

1. Insufficient task-level mathematical modeling: Existing studies are mostly based on macroscopic flexible load representations and lack a unified task-level formulation of temporal flexibility, spatial migration capability, and queue evolution. This makes it difficult to capture the combinatorial decision structure and fine-grained dynamic characteristics of heterogeneous workloads during scheduling.
2. Incomplete discrete–continuous coupling between computing and power systems: Most methods address computing scheduling and power operation separately, failing to jointly model the coupling among computing utilization, IT equipment power consumption, cooling demand, power distribution losses, energy storage operation, and task migration. As a result, the resulting optimization model may not fully describe the actual interaction between binary scheduling decisions and continuous power-side variables.
3. Lack of tailored solution methods for strongly coupled optimization: Due to the strong coupling between discrete task scheduling and continuous power decisions, existing methods mainly rely on general-purpose heuristics. These methods are not specifically designed for the hybrid mathematical structure of multi-data-center coordinated operation and often suffer from large search spaces, high evaluation costs, and unsatisfactory convergence performance.

To address the above issues, this paper formulates a multi-data-center mixed-integer nonlinear programming model (MDC-MINLP) for task-level computation–power coordinated scheduling and proposes an Adaptive Large Neighborhood Search algorithm with Hierarchical Collaboration (HC-ALNS). The proposed model provides a unified mathematical formulation for task spatiotemporal migration, device-level computing utilization, flexible queue evolution, power consumption, energy storage operation, and quality-of-service constraints. The proposed algorithm further exploits the hierarchical structure of

the problem by combining task action-space reconstruction, surrogate objective screening, accurate power-layer evaluation, and adaptive search intensity adjustment.

The main contributions of this paper are as follows:

- A task-level mathematical scheduling model is proposed to describe the dynamic scheduling characteristics of multi-type workloads. The model explicitly represents temporal shiftability, spatial migration capability, flexible queue evolution, device-level computing capacity constraints, and quality-of-service requirements, thereby transforming heterogeneous workload management into a structured mixed-integer optimization problem.
- A computation–power coupled optimization model is formulated for multi-data-center coordinated operation. Aiming to minimize the total system operating cost, the model jointly considers task scheduling constraints, computing capacity constraints, power balance constraints, IT and auxiliary power consumption, energy storage operation, and migration cost. By integrating electricity purchasing cost, IT operation loss, storage degradation cost, and migration cost into a unified objective function, the proposed MDC-MINLP captures the discrete–continuous coupling between workload allocation and power-side operation.
- An Adaptive Large Neighborhood Search algorithm with Hierarchical Collaboration (HC-ALNS) is proposed for the resulting large-scale hybrid optimization problem. The algorithm reduces combinatorial complexity through task action-space reconstruction, accelerates candidate solution screening using a surrogate objective, performs accurate power-layer evaluation for selected solutions, and adaptively adjusts search intensity according to convergence behavior. This design improves the efficiency and solution quality of the optimization process while maintaining consistency with the real total cost objective.

This study is organized into three main parts. First, a multi-data-center mixed-integer nonlinear programming model for task-level computation–power coordinated scheduling is formulated, and an Adaptive Large Neighborhood Search algorithm with Hierarchical Collaboration is developed. Second, the performance of the proposed method is evaluated through numerical experiments. Finally, the main findings are summarized and conclusions are drawn.

2. Materials and Methods

This section presents the scheduling framework, formulates a multi-data-center mixed-integer nonlinear programming model (MDC-MINLP) for task-level computation–power coordinated scheduling, and constructs the total system operating cost model. An Adaptive Large Neighborhood Search algorithm with Hierarchical Collaboration is then developed to solve the resulting problem.

2.1. Scheduling Framework

The overall framework of the studied system is illustrated in Figure 1. Traditional studies on data centers usually focus on real-time workload processing, whereas this paper models multiple data centers as integrated entities with schedulable workload resources and controllable power-side resources. During normal operation, data centers receive user workloads and execute them using computing devices supplied by the power grid, while cooling and power distribution systems provide auxiliary support. Therefore, the operational state of a data center is determined not only by the amount of workload it receives but also by task scheduling decisions, resource utilization levels, and corresponding power consumption.

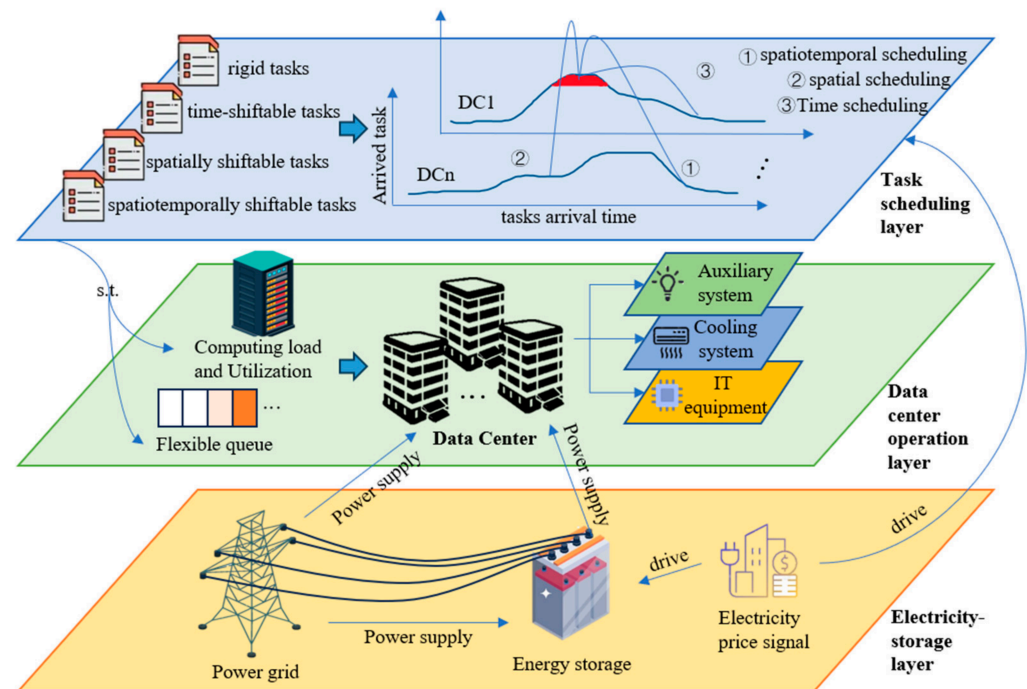


Figure 1. Framework of the coordinated operation.

Specifically, the multi-data-center system is represented as a three-layer framework consisting of the task-scheduling layer, the data center operation layer, and the power support layer. At the task-scheduling layer, workload flexibility is modeled through feasible temporal and spatial scheduling decisions. Some tasks can be shifted to different time slots, some can be migrated among data centers, and some possess both types of flexibility. These decisions determine the task execution time and location, and thus define the workload distribution over the scheduling horizon. At the data center operation layer, task scheduling results are mapped into device-level computing loads and utilization levels, which further determine IT power consumption and auxiliary power demand. At the power support layer, energy storage charging and discharging decisions are coordinated with time-of-use (TOU) electricity prices and grid procurement, providing additional degrees of freedom for cost optimization.

Therefore, the three layers are not independent modules but are linked by mathematical mappings and coupling constraints. Task scheduling decisions determine computing load and utilization; utilization determines IT, cooling, and distribution power consumption; and power demand further affects grid purchase and storage operation. Conversely, electricity price parameters and storage constraints influence the optimal spatiotemporal allocation of workloads through the total cost objective. This bidirectional relationship forms a closed-loop computation–power coordinated scheduling mechanism.

These interactions indicate the necessity of establishing a unified MDC-MINLP for multi-data-center operation. Such a model should jointly describe task scheduling constraints, computing capacity constraints, power balance constraints, storage operation constraints, and quality-of-service requirements, thereby supporting coordinated optimization of workload allocation and power-side operation.

2.2. Problem Formulation

2.2.1. Task Description and Scheduling Variables

In the multi-data-center collaborative scheduling scenario, both the execution time and execution location of a task must be determined when it enters the system. Due to the

differences in temporal and spatial scheduling flexibility among tasks, execution decisions are constrained not only by resource capacity but also by their transferability.

Let the scheduling horizon be discretized into a set of time slots denoted by $\mathcal{T}$. The set of data centers participating in collaborative scheduling is denoted by $\mathcal{N}$, and the set of tasks arriving within the scheduling horizon is labeled $\mathcal{M}$. Furthermore, let $\mathcal{I}$ represent the set of all computing devices in the system and let $\mathcal{I}_k \subseteq \mathcal{I}$ indicate the set of devices from data center k . For any task $m \in \mathcal{M}$, let a_m denote its arrival time slot, let o_m represent its original data center, let p_m indicate its processing duration, and let c_m represent its computing demand. In addition, $\delta_m^T \in \{0,1\}$ and $\delta_m^S \in \{0,1\}$ will indicate whether task m can be temporally shifted and spatially migrated, respectively. The basic scheduling principles are illustrated in Figure 2.

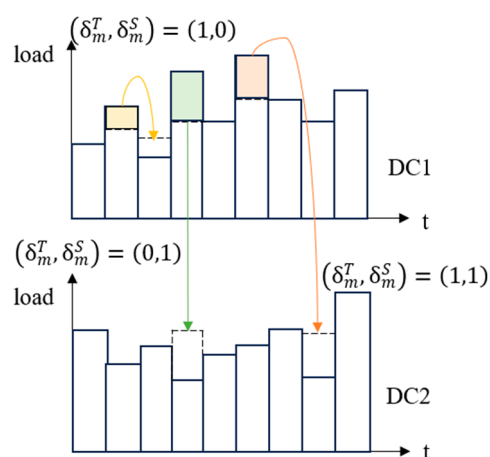


Figure 2. Flexible response principle for tasks.

To characterize the execution device and the starting time of tasks, a binary decision variable $x_{m,i,t}$ is introduced. If task m is initiated for execution on device i at time slot t , then $x_{m,i,t} = 1$; otherwise, $x_{m,i,t} = 0$.

Since each task can only be initiated once within the scheduling horizon, the following constraint holds:

$$\sum_{i \in \mathcal{I}} \sum_{t \in \mathcal{T}} x_{m,i,t} = 1, \quad \forall m \in \mathcal{M}, \quad (1)$$

during the task-scheduling process, it is necessary to satisfy the fundamental temporal constraint that tasks cannot be executed before their arrival. Scheduling flexibility constraints must also be considered. Specifically, when a task cannot be temporally shifted, its execution time is fixed at its arrival time, whereas when a task cannot be spatially migrated, its execution location is restricted to the set of devices $\mathcal{I}_{o_m}$ within its original data center o_m . These constraints can be uniformly expressed as follows:

$$\begin{cases} x_{m,i,t} = 0, & \forall m \in \mathcal{M}, i \in \mathcal{I}, t < a_m, \\ x_{m,i,t} = 0, & \forall m \in \mathcal{M}, i \in \mathcal{I}, t \in \mathcal{T}, \delta_m^T = 0, t \neq a_m, \\ x_{m,i,t} = 0, & \forall m \in \mathcal{M}, i \in \mathcal{I} \setminus \mathcal{I}_{o_m}, t \in \mathcal{T}, \delta_m^S = 0. \end{cases} \quad (2)$$

2.2.2. Data Center Computing Load Model

After its initiation, a task continuously occupies computing resources during its execution period. Therefore, the computing load of a device at any time slot is jointly determined by all ongoing tasks. Let $L_{i,t}$ denote the computing load of device i at time slot t , which is written as follows:

$$L_{i,t} = \sum_{m \in \mathcal{M}} \sum_{\tau \leq t} c_m x_{m,i,\tau} \mathbf{1}(t \in [\tau, \tau + p_m]), \quad \forall i \in \mathcal{I}, t \in \mathcal{T}, \quad (3)$$

where $\mathbf{1}(\cdot)$ denotes the indicator function. When task m is initiated at time slot τ and remains in execution at time slot t , it contributes c_m to the computing load of device i at time slot t .

Let $C_{i,t}$ denote the available computing capacity of device i at time slot t , then:

$$L_{i,t} \leq C_{i,t}, \quad \forall i \in \mathcal{I}, t \in \mathcal{T}. \quad (4)$$

Furthermore, the total computing load of data center k at time slot t is obtained by aggregating at the device level:

$$L_{k,t} = \sum_{i \in \mathcal{I}_k} L_{i,t}, \quad \forall k \in \mathcal{N}, t \in \mathcal{T}. \quad (5)$$

2.2.3. Flexible Task Queue Evolution Model

In a multi-type workload scenario, only tasks that allow temporal shifting remain intact across time slots. Therefore, the queue in the system should only be defined for tasks with temporal flexibility. Based on the task-level scheduling variables, let $Q_{i,t}$ denote the flexible waiting queue of device i at the end of time slot t , which consists of temporally shiftable tasks that have been assigned to device i , and have already arrived, but have not yet been initiated. It can be expressed as follows:

$$Q_{i,t} = \sum_{m \in \mathcal{M}, \delta_m^T=1} c_m \sum_{t=a_m}^t \sum_{t+1}^T x_{m,i,t+1}, \quad \forall i \in \mathcal{I}, t \in \mathcal{T}. \quad (6)$$

In the above expression, the inner summation indicates that task m has already arrived before time slot t , while its initiation time is scheduled after t . Therefore, the task remains in the waiting state at the end of time slot t .

To describe the queue variation between adjacent time slots, let $A_{i,t}^Q$ represent the workload entering the flexible queue of device i at time slot t and let $D_{i,t}^Q$ denote the workload leaving the flexible queue at time slot t due to task initiation. Then,

$$Q_{i,t+1} = Q_{i,t} + A_{i,t}^Q - D_{i,t}^Q, \quad \forall i \in \mathcal{I}, t \in \mathcal{T}, \quad (7)$$

where

$$A_{i,t}^Q = \sum_{m \in \mathcal{M}, \delta_m^T=1, a_m=t} c_m \sum_{t+1}^T x_{m,i,t+1}, \quad \forall i \in \mathcal{I}, t \in \mathcal{T}, \quad (8)$$

$$D_{i,t}^Q = \sum_{m \in \mathcal{M}, \delta_m^T=1, a_m < t} c_m x_{m,i,t}, \quad \forall i \in \mathcal{I}, t \in \mathcal{T}, \quad (9)$$

$A_{i,t}^Q$ represents the workload of temporally shiftable tasks that arrive at time slot t but are not executed immediately and are therefore deferred to later time slots. $D_{i,t}^Q$ represents the workload of tasks that have previously entered the flexible queue and are initiated at time slot t .

2.2.4. QoS Guarantee Model

Allowing temporal shifting does not imply that tasks can be delayed indefinitely. Constraints must still be imposed on the waiting behavior of flexible tasks. Meanwhile, to limit the accumulation of flexible tasks at the system level, constraints on the size of the

flexible queue for each device are also required. To this end, the startup delay of task m is first defined as:

$$W_m = \sum_{i \in \mathcal{I}} \sum_{t \in \mathcal{T}} (t - a_m) x_{m,i,t}, \quad \forall m \in \mathcal{M}, \quad (10)$$

where W_m denotes the waiting time from arrival to execution initiation. For tasks that do not permit temporal shifting, $W_m = 0$; for tasks that do allow temporal shifting, the waiting time cannot exceed the upper bound permitted by quality-of-service requirements. These QoS requirements can be uniformly expressed as follows:

$$\begin{cases} W_m \leq \bar{W}_m, & \forall m \in \mathcal{M}, \delta_m^T = 1, \\ 0 \leq Q_{i,t} \leq \bar{Q}_i, & \forall i \in \mathcal{I}, t \in \mathcal{T}, \\ Q_{i,T} = Q_{i,0}, & \forall i \in \mathcal{I}, \end{cases} \quad (11)$$

where $\bar{W}_m$ denotes the maximum tolerable waiting delay of task m and $\bar{Q}_i$ denotes the maximum tolerable flexible queue capacity of device i . The above constraints guarantee QoS based on two aspects. On the one hand, they prevent excessive delay of individual tasks by limiting the task waiting time. They also prevent the continuous accumulation of deferrable workloads in the system by restricting the size of the flexible queue, while the cyclic consistency condition ensures operational stability across scheduling periods.

2.2.5. IT Equipment Power Model

The task scheduling results determine the computing resource utilization of each device at each time slot, which further affects the IT equipment power consumption. To characterize the relationship between computing load and IT power, the computing utilization variable $\rho_{i,t}$ is introduced, representing the utilization level of device i at time slot t , defined as:

$$\rho_{i,t} = \frac{L_{i,t}}{C_{i,t}}, \quad \forall i \in \mathcal{I}, \forall t \in \mathcal{T}, \quad (12)$$

$$0 \leq \rho_{i,t} \leq 1, \quad \forall i \in \mathcal{I}, \forall t \in \mathcal{T}, \quad (13)$$

where $L_{i,t}$ denotes the computing load of device i at time slot t and $C_{i,t}$ denotes the available computing capacity of device i at that time slot.

Let $P_{i,t}^{IT}$ denote the IT power of device i at time slot t , let P_i^{idle} denote its idle power, and let P_i^{full} denote its full-load power. The IT equipment power model is given by

$$P_{i,t}^{IT} = P_i^{idle} + (P_i^{full} - P_i^{idle}) \rho_{i,t}^{\tau_i}, \quad \forall i \in \mathcal{I}, \forall t \in \mathcal{T}, \quad (14)$$

where τ_i is the nonlinear power coefficient of device i , which is used to characterize the response of IT power to changes in computing utilization [21].

Furthermore, the total IT power of data center k at time slot t is

$$P_{k,t}^{IT} = \sum_{i \in \mathcal{I}_k} P_{i,t}^{IT}, \quad \forall k \in \mathcal{N}, t \in \mathcal{T}. \quad (15)$$

2.2.6. Auxiliary System Power and PUE Model

In addition to IT equipment, data centers also require cooling systems, power supply and distribution systems, and other auxiliary facilities, whose energy consumption is coupled with the IT load. Let $P_{k,t}^{cool}$ denote the cooling system power of data center k at time slot t and let $P_{k,t}^{loss}$ represent the power loss of the supply and distribution system. The auxiliary system power is modeled as follows:

$$P_{k,t}^{cool} = P_k^{cool,0} + \alpha_k^{cool} P_{k,t}^{IT}, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}, \quad (16)$$

$$P_{k,t}^{loss} = P_k^{loss,0} + \alpha_k^{loss} (P_{k,t}^{IT} + P_{k,t}^{cool}), \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}. \quad (17)$$

The total power load of data center k at time slot t is:

$$P_{k,t}^{load} = P_{k,t}^{IT} + P_{k,t}^{cool} + P_{k,t}^{loss}, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}. \quad (18)$$

To characterize the overall energy efficiency of the data center, the power usage effectiveness is introduced:

$$PUE_{k,t} = \frac{P_{k,t}^{load}}{P_{k,t}^{IT}}, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}, \quad (19)$$

$$PUE_{k,t} \geq 1, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}, \quad (20)$$

where $PUE_{k,t}$ represents the ratio between total power consumption and IT power consumption.

2.2.7. Energy Storage System Model

To enhance the power regulation capability of data centers in the temporal dimension, an energy storage system is introduced on the power side. The energy storage system can charge during low-load periods and discharge during high-load periods, thereby smoothing the grid power and improving operational flexibility. Let $SOC_{k,t}$ denote the state of charge of data center k at time slot t ; let $P_{k,t}^{ch}$ and $P_{k,t}^{dis}$ denote the charging and discharging power, respectively; and let Δt denote the duration of a time slot. The energy storage dynamics are given by:

$$SOC_{k,t+1} = SOC_{k,t} + \frac{\eta_k^{ch} P_{k,t}^{ch} \Delta t}{E_k^{bat}} - \frac{P_{k,t}^{dis} \Delta t}{\eta_k^{dis} E_k^{bat}}, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}, \quad (21)$$

$$SOC_k^{min} \leq SOC_{k,t} \leq SOC_k^{max}, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}, \quad (22)$$

$$SOC_{k,T} = SOC_{k,0}, \quad \forall k \in \mathcal{N}. \quad (23)$$

To prevent simultaneous charging and discharging within the same time slot, a binary variable $y_{k,t}$ is introduced:

$$0 \leq P_{k,t}^{ch} \leq y_{k,t} P_k^{ch,max}, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}, \quad (24)$$

$$0 \leq P_{k,t}^{dis} \leq (1 - y_{k,t}) P_k^{dis,max}, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}, \quad (25)$$

where E_k^{bat} denotes the rated capacity of the energy storage system and η_k^{ch} and η_k^{dis} denote the charging and discharging efficiencies, respectively. The above cyclic condition ensures that the energy storage system operates consistently at the beginning and end of the scheduling horizon.

2.2.8. Grid Power Supply Model

The data centers are modeled as large adjustable load-side units connected to the external grid at the point of common coupling (PCC) on the distribution side. Within the adopted single-bus equivalent power-supply framework, the IT equipment, cooling system, and power supply and distribution losses jointly constitute the total data center load. For each time slot, this load is supplied by grid purchasing power and energy storage discharging power, while energy storage charging is treated as additional power demand connected to the same bus. Accordingly, the corresponding single-bus power balance relationship is established to describe the coordinated operation of data center load, energy storage, and external grid power procurement.

Let $P_{k,t}^{grid}$ denote the grid purchasing power of data center k at time slot t , then

$$P_{k,t}^{grid} + P_{k,t}^{dis} = P_{k,t}^{load} + P_{k,t}^{ch}, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}, \quad (26)$$

$$0 \leq P_{k,t}^{grid} \leq P_k^{grid,max}, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}. \quad (27)$$

To facilitate subsequent electricity pricing settlement and energy accounting, time slot-level energy variables are introduced:

$$E_{k,t}^{grid} = P_{k,t}^{grid} \Delta t, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}, \quad (28)$$

$$E_{k,t}^{load} = P_{k,t}^{load} \Delta t, \quad \forall k \in \mathcal{N}, \forall t \in \mathcal{T}, \quad (29)$$

where $E_{k,t}^{grid}$ represents the total amount of electricity purchased from the grid by data center k at time slot t and $E_{k,t}^{load}$ represents the total power consumption of data center k at time slot t . Both are obtained by converting power to energy based on Δt .

2.2.9. Single-Data-Center Operating Cost Model

Let F_k denote the total operating cost of data center k during the scheduling horizon, then

$$F_k = F_k^{grid} + F_k^{IT} + F_k^{bat} + F_k^{mig}, \quad \forall k \in \mathcal{N}, \quad (30)$$

where F_k^{grid} represents the grid electricity purchasing cost, F_k^{IT} represents the IT equipment operation loss cost, F_k^{bat} represents the energy storage system operation loss cost, and F_k^{mig} represents the task migration loss cost incurred by executing tasks from other data centers.

The grid electricity purchasing cost is determined by the purchased energy and the corresponding electricity price at each time slot, which we express as follows:

$$F_k^{grid} = \sum_{t \in \mathcal{T}} \lambda_{k,t} E_{k,t}^{grid}, \quad \forall k \in \mathcal{N}, \quad (31)$$

where $\lambda_{k,t}$ denotes the electricity price of data center k at time slot t .

During long-term operation, IT equipment incurs hardware degradation, which is closely related to computing utilization. When the utilization deviates from the optimal operating range, the degradation increases. Let λ_{IT} denote the unit computing operation loss cost coefficient, let γ denote the amplification coefficient of utilization deviation, and let ρ_{opt} denote the optimal utilization level. The IT equipment operation loss cost of data center k is expressed as follows:

$$F_k^{IT} = \lambda_{IT} \sum_{i \in \mathcal{I}_k} \sum_{t \in \mathcal{T}} C_{i,t} (\rho_{i,t} + \gamma |\rho_{i,t} - \rho_{opt}|) \Delta t, \quad \forall k \in \mathcal{N}, \quad (32)$$

the above expression is directly constructed based on the device utilization $\rho_{i,t}$ and available computing capacity $C_{i,t}$, reflecting the IT equipment operation loss under different load levels.

During charging and discharging, the energy storage system incurs battery degradation and lifetime loss. Considering the cumulative impact of these processes, let λ_{bat} denote the lifecycle loss cost per unit energy throughput. The energy storage system operation loss cost is expressed as follows:

$$F_k^{bat} = \lambda_{bat} \left(\sum_{t \in \mathcal{T}} P_{k,t}^{ch} \Delta t + \frac{1}{\eta_k^{dis}} \sum_{t \in \mathcal{T}} P_{k,t}^{dis} \Delta t \right), \quad \forall k \in \mathcal{N}. \quad (33)$$

When tasks are executed in data centers different from their original locations, additional costs arise as a result of data transfer, coordination, and migration. Let λ_{mig} denote the migration loss cost coefficient per unit computing demand. The migration loss cost borne by data center k is expressed as follows:

$$F_k^{mig} = \lambda_{mig} \sum_{m \in \mathcal{M}} \sum_{i \in \mathcal{I}_k} \sum_{t \in \mathcal{T}} c_m \mathbf{1}(k \neq o_m) x_{m,i,t}, \quad \forall k \in \mathcal{N}. \quad (34)$$

2.2.10. Overall Objective Function of Multi-Data-Center System

In the multi-data-center collaborative scheduling scenario, the overall objective is to minimize the total operating cost of all data centers during the scheduling horizon. The objective function is given by:

$$\min F = \sum_{k \in \mathcal{N}} F_k. \quad (35)$$

Expanding the single-data-center cost yields

$$\min F = \sum_{k \in \mathcal{N}} (F_k^{grid} + F_k^{IT} + F_k^{bat} + F_k^{mig}), \quad (36)$$

this objective function takes data centers as cost aggregation units and integrates task-scheduling decisions and power operation states into a unified optimization framework, enabling coordinated economic optimization of computing and power resources under service quality and system operational safety constraints. These are presented as the formulations in Section 2.2.

2.3. Solution Algorithm

The multi-data-center collaborative scheduling problem involves both discrete decisions, such as task execution location and start time, and continuous decisions, such as grid purchasing power, energy storage power, and state of charge. Therefore, it is a discrete–continuous coupled hybrid optimization problem. To improve the solution efficiency in large-scale scenarios, an Adaptive Large Neighborhood Search algorithm with Hierarchical Collaboration (HC-ALNS) is developed. The algorithm iteratively solves the real total cost minimization problem through “task-layer discrete reconstruction—power-layer accurate evaluation—adaptive search intensity adjustment”.

2.3.1. Task Action Space Reconstruction

For any task $m \in \mathcal{M}$, its scheduling result is reconstructed as an action pair:

$$b_m = (i_m, t_m), \quad (37)$$

where $i_m \in \mathcal{I}$ denotes the execution device of task m and $t_m \in \mathcal{T}$ denotes the start time slot of task m . Accordingly, the scheduling solution of the system can be expressed as follows:

$$\mathbf{b} = \{b_m \mid m \in \mathcal{M}\}. \quad (38)$$

Considering the temporal transferability δ_m^T , spatial transferability δ_m^S , original data center o_m , arrival time slot a_m , and the maximum tolerable waiting delay $\overline{W}_m$, the candidate action set of task m is defined as shown below:

$$\mathcal{A}_m = \left\{ (i, t) \mid \delta_m^S = 0 \Rightarrow i \in \mathcal{I}_{o_m}, \delta_m^T = 0 \Rightarrow t = a_m, 0 \leq t - a_m \leq \overline{W}_m \right\}. \quad (39)$$

To enhance the scheduling effect, a cross-day segmented scheduling mechanism is adopted. Through this mechanism, the time slot variable t can be extended, in the sense

of cyclic time slots, to the allowable forward-shifting window. Accordingly, the original problem is transformed into a combinatorial optimization problem over the candidate action set $\mathcal{A}_m$.

2.3.2. Surrogate Objective Construction

To reduce the computational overhead caused by frequent real evaluations, a surrogate objective that is consistent with the variation direction of the real total cost is constructed:

$$\tilde{F}(\mathbf{b}) = \sum_{k \in \mathcal{N}} \tilde{F}_k(\mathbf{b}) + \tilde{F}^{move}(\mathbf{b}), \quad (40)$$

where $\tilde{F}(\mathbf{b})$ denotes the surrogate operating cost of data center k and $\tilde{F}^{move}(\mathbf{b})$ denotes the additional cost caused by task spatiotemporal shifts.

For any data center k , define

$$\tilde{F}_k = \omega_1 \tilde{F}_k^{grid} + \omega_2 \tilde{F}_k^{peak} + \omega_3 \tilde{F}_k^{smooth} + \omega_4 \tilde{F}_k^{util} + \omega_5 \tilde{F}_k^{bat} + \omega_6 \tilde{F}_k^{cap}, \quad (41)$$

where ω_i represents the weighting coefficients of each surrogate term. Each term is defined as follows:

$$\tilde{F}_k^{grid} = \sum_{t \in \mathcal{T}} \lambda_{k,t} P_{k,t}^{load} \Delta t, \quad (42)$$

$$\tilde{F}_k^{peak} = \sum_{i \in \mathcal{I}_k} \sum_{t \in \mathcal{T}} \left[\alpha_1 (\rho_{i,t} - \rho^{warn})_+^2 + \alpha_2 (\rho_{i,t} - \rho^{max})_+^4 \right], \quad (43)$$

$$\tilde{F}_k^{smooth} = \alpha_3 \sum_{i \in \mathcal{I}_k} \sum_{t=1}^{T-1} |\rho_{i,t+1} - \rho_{i,t}|, \quad (44)$$

$$\tilde{F}_k^{util} = \alpha_4 \sum_{i \in \mathcal{I}_k} \sum_{t \in \mathcal{T}} |\rho_{i,t} - \rho_{opt}|, \quad (45)$$

$$\tilde{F}_k^{bat} = \alpha_5 \sum_{t \in \mathcal{T}} (\lambda_{k,t} - \bar{\lambda})_+ P_{k,t}^{load} \Delta t, \quad (46)$$

$$\tilde{F}_k^{cap} = \alpha_6 \sum_{i \in \mathcal{I}_k} \sum_{t \in \mathcal{T}} (\rho_{i,t} - \rho^{max})_+, \quad (47)$$

where ρ^{warn} denotes the utilization warning threshold, ρ_{opt} denotes the desired optimal utilization level, and $\bar{\lambda}$ denotes the average electricity price over the scheduling horizon; the notation $(x)_+ = \max(x, 0)$ denotes the positive-part function.

The additional cost caused by task spatiotemporal shifts is expressed as follows:

$$\tilde{F}_k^{move} = \sum_{m \in \mathcal{M}} \left[\lambda_{mig} c_m \mathbf{1}(i_m \notin \mathcal{I}_{o_m}) + \lambda^{shift} |t_m - a_m| \right], \quad (48)$$

where λ^{shift} denotes the penalty coefficient per unit time-slot shift; the surrogate objective is only used for search screening, while the final quality of the candidate solutions is determined by the real total cost.

2.3.3. High-Impact Task Selection

To improve search efficiency, a comprehensive impact score is defined for tasks in the current solution:

$$\Phi_m = (c_m p_m) \psi_m^{flex} \psi_m^{cong} \psi_m^{price}, \quad (49)$$

where $c_m p_m$ denotes the task size and ψ_m^{flex} , ψ_m^{cong} , and ψ_m^{price} denote the flexibility coefficient, congestion sensitivity coefficient, and price sensitivity coefficient, respectively. Their specific forms are:

$$\psi_m^{flex} = 1 + \beta_1 \delta_m^T + \beta_2 \delta_m^S, \quad (50)$$

$$\psi_m^{cong} = 1 + \beta_3 \bar{\rho}_m + \beta_4 (\rho_m^{peak} - \rho^{warn})_+, \quad (51)$$

$$\psi_m^{price} = 1 + \beta_5 \Delta \lambda_m, \quad (52)$$

where β_i represents the weighting parameters; $\bar{\rho}_m$ denotes the average utilization over the current execution interval of task m ; ρ_m^{peak} denotes the peak utilization over this interval; and $\Delta \lambda_m$ denotes the difference between the average electricity price over the current execution interval and that over the candidate low-price interval. Tasks are sorted in descending order of Φ_m , and the top-ranked tasks are selected to form the core task pool, which is then combined with a small number of randomly selected movable tasks to create the search pool of the current iteration.

2.3.4. Initial Feasible Solution Construction

Define the task size indicator as follows:

$$s_m = c_m p_m. \quad (53)$$

Tasks are placed sequentially in descending order of s_m . For any task m , feasible actions are initially searched for within a reduced candidate set; if unsuccessful, the search is expanded to the full action set $\mathcal{A}_m$. For a candidate action (i, t) , the surrogate incremental cost is defined as follows:

$$\Delta \tilde{F}_{m,i,t} = \tilde{F}(\mathbf{b} \oplus (m, i, t)) - \tilde{F}(\mathbf{b}), \quad (54)$$

where $\mathbf{b} \oplus (m, i, t)$ denotes the new scheduling solution obtained by assigning task m to action (i, t) . Then, select

$$(i_m^*, t_m^*) = \arg \min_{(i,t) \in \mathcal{A}_m^{feas}} \Delta \tilde{F}_{m,i,t}, \quad (55)$$

where $\mathcal{A}_m^{feas} \subseteq \mathcal{A}_m$ denotes the feasible action set satisfying the hard utilization constraint.

If the task still cannot be directly inserted, a position with the minimal violation degree is selected for temporary placement. Then, through a directed repair process, movable tasks covering overloaded time slots are rearranged, resulting in a feasible solution satisfying

$$\rho_{i,t} \leq \rho^{max}, \quad \forall i \in \mathcal{I}, t \in \mathcal{T}. \quad (56)$$

2.3.5. Large Neighborhood Destroy–Repair Reconstruction

Let the current solution at iteration r be $\mathbf{b}^{(r)}$ and select a batch of tasks $\mathcal{B}^{(r)}$ from the search pool. First, all tasks $m \in \mathcal{B}^{(r)}$ are removed from the current solution to obtain a partial solution $\mathbf{b}^{(r),-}$; then tasks are reinserted in descending order of task size.

For any task m to be inserted, solve

$$(i_m^*, t_m^*) = \arg \min_{(i,t) \in \mathcal{A}_m^{feas}} \Delta \tilde{F}_{m,i,t}^{(r)}, \quad (57)$$

where

$$\Delta \tilde{F}_{m,i,t}^{(r)} = \tilde{F}(\mathbf{b}^{(r),-} \oplus (m, i, t)) - \tilde{F}(\mathbf{b}^{(r),-}). \quad (58)$$

To enhance search diversity, instead of always selecting the single best action, one candidate is randomly selected from the top q near-optimal candidates:

$$(i_m^*, t_m^*) \sim \text{Unif}\left(\text{Top-}q\left\{\Delta\tilde{F}_{m,i,t}^{(r)}\right\}\right), \quad (59)$$

where $\text{Top-}q\{\cdot\}$ denotes the set of the top q candidates sorted in ascending order of surrogate incremental cost and $\text{Unif}(\cdot)$ denotes uniform random selection from this set. When $q = 1$, it degenerates into pure greedy repair.

2.3.6. Power-Layer Accurate Evaluation

For any candidate solution $\mathbf{b}$ generated at the task layer, the time slot-level computing load $L_{i,t}$, utilization $\rho_{i,t}$, and total power load $P_{k,t}^{\text{load}}$ are determined. On this basis, for each data center k , the energy storage operation subproblem is solved:

$$\min \sum_{t \in \mathcal{T}} \left[\lambda_{k,t} E_{k,t}^{\text{grid}} + \lambda_{\text{bat}} \left(P_{k,t}^{\text{ch}} \Delta t + \frac{1}{\eta_k^{\text{dis}}} P_{k,t}^{\text{dis}} \Delta t \right) \right] + \chi_k^{\text{term}}, \quad (60)$$

where χ_k^{term} denotes the penalty term for deviation of the terminal state of charge.

Furthermore, the real total cost of a candidate solution is expressed as follows:

$$F(\mathbf{b}) = \sum_{k \in \mathcal{N}} \left(F_k^{\text{grid}} + F_k^{\text{IT}} + F_k^{\text{bat}} + F_k^{\text{mig}} \right). \quad (61)$$

The algorithm first uses $\tilde{F}$ to screen candidate solutions and then performs accurate evaluation using the real objective F for only a small number of high-quality candidates. This allows the acceptance of the candidate solution to be determined.

2.3.7. Single-Task Refinement and Stagnation Enhancement

In addition to batch large neighborhood reconstruction, a single-task refinement operator is introduced. For a selected task m , a “remove–reinsert–evaluate” process is performed. If the adjusted candidate solution $\hat{\mathbf{b}}$ satisfies

$$F(\hat{\mathbf{b}}) < F(\mathbf{b}), \quad (62)$$

the modification is accepted.

When the algorithm exhibits insufficient improvement over consecutive iterations, a stagnation enhancement mechanism is triggered. Increasing the destroy ratio, enlarging the number of perturbed tasks, and increasing the depth of random candidate selection enhances the structural reconstruction capability. Let the current solution be $\mathbf{b}^{(r)}$ while the perturbed candidate solution is $\hat{\mathbf{b}}^{(r)}$. Slight controlled uphill acceptance is allowed at specific iterations:

$$F(\hat{\mathbf{b}}^{(r)}) \leq F(\mathbf{b}^{(r)}) + \varepsilon_r^{\text{up}}, \quad (63)$$

where $\varepsilon_r^{\text{up}}$ denotes the uphill tolerance threshold at iteration r .

2.3.8. Adaptive Search Intensity Adjustment

Let $\bar{\eta}^{(r)}$ denote the average relative improvement over the preceding several iterations before r , $u^{(r)}$ denote the number of consecutive stagnation iterations, and let $\zeta^{(r)}$ denote the iteration progress ratio. The search intensity indicator is defined as follows:

$$\Gamma^{(r)} = 1 + \mu_1 \zeta^{(r)} + \mu_2 u^{(r)} + \mu_3 \mathbf{1}(\bar{\eta}^{(r)} < \eta_{\min}), \quad (64)$$

where μ_i are adaptive adjustment parameters and $\eta_{\min}$ is the minimum effective improvement threshold.

According to $\Gamma^{(r)}$, the task pool size $M_{pool}^{(r)}$, batch size $B^{(r)}$, real evaluation budget $E^{(r)}$, number of single-task refinements $N_{fine}^{(r)}$, and perturbation rounds $R_{shake}^{(r)}$ are dynamically adjusted as follows:

$$M_{pool}^{(r)} = \mathcal{G}_1(\Gamma^{(r)}), B^{(r)} = \mathcal{G}_2(\Gamma^{(r)}), E^{(r)} = \mathcal{G}_3(\Gamma^{(r)}), N_{fine}^{(r)} = \mathcal{G}_4(\Gamma^{(r)}), R_{shake}^{(r)} = \mathcal{G}_5(\Gamma^{(r)}), \quad (65)$$

where $\mathcal{G}_i(\cdot)$ denotes bounded mapping functions from the search intensity indicator to the corresponding control parameters. In this way, a dynamic balance between global exploration capability and local convergence efficiency is achieved.

2.3.9. Convergence Criterion

Let the current solution cost and the historical best cost at iteration r be $F^{cur,(r)}$ and $F^{best,(r)}$, respectively, then:

$$F^{best,(r+1)} \leq F^{best,(r)}. \quad (66)$$

The algorithm terminates when any of the following conditions is satisfied:

$$r \geq R^{max}, \quad (67)$$

$$\frac{F^{best,(r-h)} - F^{best,(r)}}{F^{best,(0)}} \leq \varepsilon, \quad (68)$$

where R^{max} is the maximum number of iterations, h is the sliding window length, and ε is the minimum improvement threshold.

Finally, the historical best scheduling solution is output as follows:

$$b^* = \arg \min_r F^{best,(r)}. \quad (69)$$

The overall flow of the algorithm is shown in Figure 3:

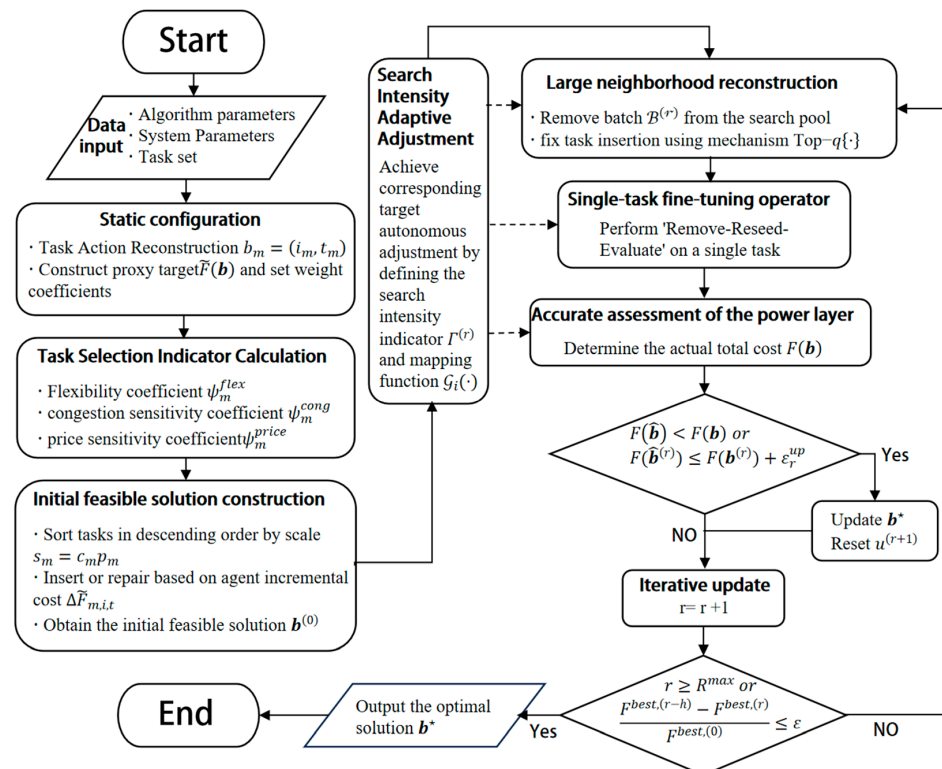


Figure 3. Workflow of the proposed HC-ALNS algorithm.

3. Results

This section presents the case study results and corresponding analyses. The experimental setup and scenario settings are first introduced, followed by a comprehensive evaluation of model effectiveness under different scheduling strategies. Subsequently, the performance of the proposed algorithm is assessed through comparative analysis. Finally, sensitivity and robustness analyses are conducted to further examine the impact of key parameters on system performance.

3.1. Experimental Setup

To validate the effectiveness of the proposed model, we have constructed a coordinated computing–power scheduling system consisting of three geographically distributed data centers in Tianjin, Beijing, and Hebei, respectively. Heterogeneity is considered across data centers in terms of both computing resources and power supply capabilities. The corresponding parameters are summarized in Table 1. Detailed parameters are listed in Tables A1 and A2 of Appendix A. The workload arrival profiles of the three data centers are derived from the Alibaba Cluster Trace [22], which serves as a real-world workload input. Specifically, task data from a single day are extracted, and the original instance-level records are preprocessed through data cleaning and aggregation to obtain 30,637 task-level inputs. Furthermore, the initial task allocation across data centers is determined based on the regional electricity consumption characteristics reported in [23], reflecting realistic load distribution patterns.

Table 1. Heterogeneous parameters of a multi-data-center system.

Parameter	Symbol	DC1	DC2	DC3	Unit
Computing capacity	C_k	20,000	15,000	9000	CPU
Flexible queue capacity	$\bar{Q}_k$	4300	3000	2000	CPU
Idle IT power	p_k^{idle}	13,000	10,200	6700	kW
Peak IT power	p_k^{full}	18,200	14,950	10,200	kW
Base cooling power	$p_k^{cool,0}$	1200	950	650	kW
Base distribution loss	$p_k^{loss,0}$	380	300	210	kW
Cooling coefficient	α_k^{cool}	0.24	0.22	0.2	—
Battery capacity	E_k^{max}	20,000	16,000	12,000	kWh
Max charging power	$p_k^{ch,max}$	3000	2500	1800	kW
Max discharging power	$p_k^{dis,max}$	3000	2500	1800	kW
Grid power limit	$p_k^{grid,max}$	29,000	21,500	15,000	kW
Migration cost coefficient	γ_k	0.2	0.18	0.16	—

For the three geographically distributed regions, the corresponding time-of-use (TOU) electricity prices are collected individually [24,25]. The TOU price profiles for the three regions in 2025 are illustrated in Figure 4. To capture the intra-day workload variation, the scheduling horizon is discretized into ninety-six 15 min time slots. Meanwhile, a cross-day scheduling mechanism is introduced by defining a time partition point, allowing some tasks to be logically shifted forward to earlier low-price periods.

As shown in Figure 5, the total workload over a 24 h period exhibits significant intra-day fluctuation, ranging from 28,425.24 CPU to 33,922.81 CPU, with a peak-to-valley difference of 19.34%. The workload remains at a relatively low level during the early morning hours, then increases steadily, reaching its peak in the afternoon, followed by a gradual decline. From the perspective of workload composition, time-shiftable tasks and spatiotemporally shiftable tasks are most prevalent, accounting for 40.87% and 30.47% of the workload on average, with a combined proportion of 71.33%. In contrast, non-shiftable

tasks account for 27.12% of the workload, while tasks that are only spatially shiftable constitute a relatively small fraction of 1.54%. Overall, most workloads exhibit scheduling flexibility, with a pronounced load concentration observed during peak periods.

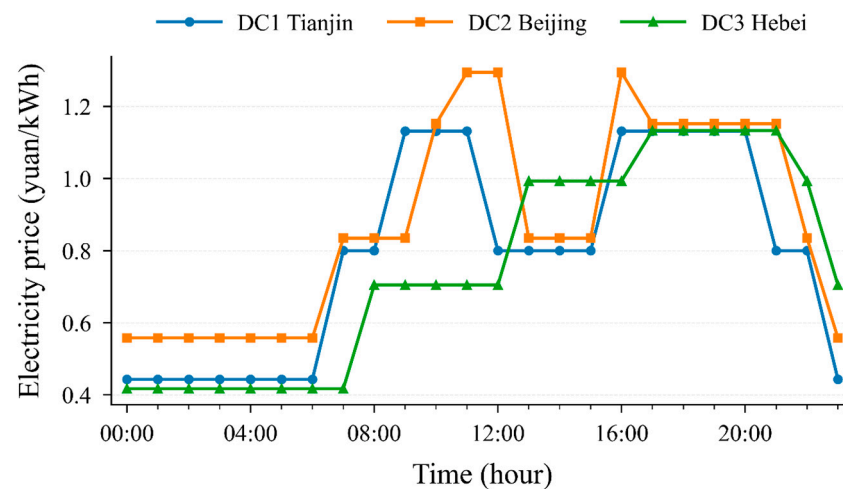


Figure 4. The electricity prices of three data centers.

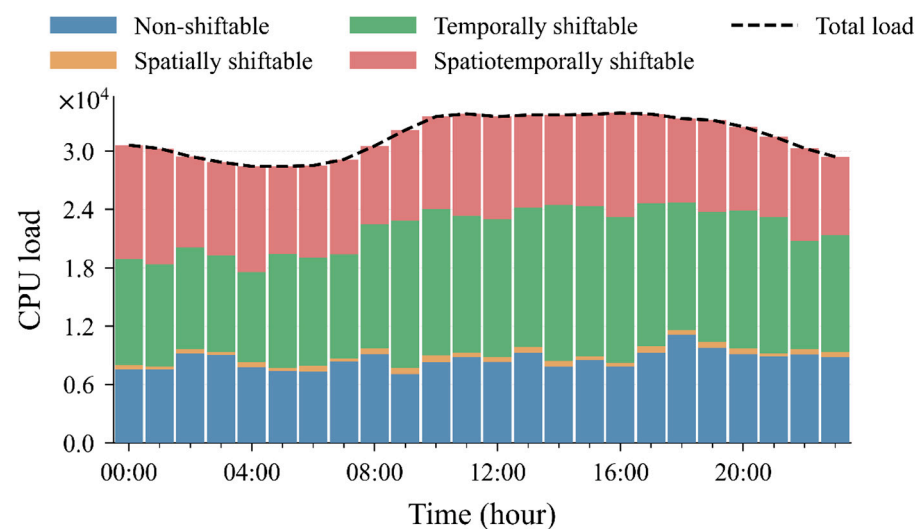


Figure 5. Twenty-four-hour aggregated load distribution.

3.2. Scenario Settings

Based on the proposed computing–power coordinated scheduling model, multiple operational scenarios are designed to evaluate the effectiveness of different scheduling strategies:

1. Scenario 1 (Baseline): No scheduling optimization is applied. All tasks are executed at their original arrival times and within their originating data centers.
2. Scenario 2 (Time-only scheduling): Only temporal flexibility is considered. Tasks are allowed to shift across time slots, but cross-data-center migration is not permitted.
3. Scenario 3 (Space-only scheduling): Only spatial flexibility is considered. Tasks can migrate across data centers while their execution time remains unchanged.
4. Scenario 4 (Spatiotemporal scheduling): Both temporal and spatial flexibility are considered. Tasks can be shifted across time slots and migrated across data centers.
5. Scenario 5 (Spatiotemporal scheduling with energy storage): Based on Scenario 4, energy storage is further integrated into system operation.

3.3. Model Effectiveness Analysis

The total system cost after optimization and the corresponding convergence processes for Scenarios 1–5 are illustrated in Figure 6. All scenarios converge to stable objective values after 49 iterations, while the final costs differ significantly under different scheduling strategies.

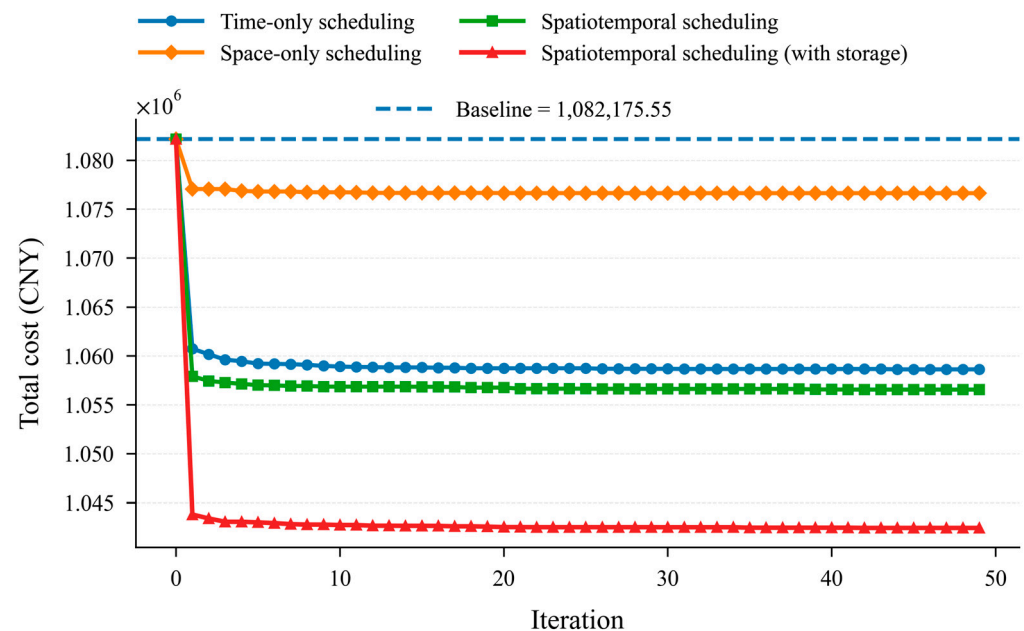


Figure 6. Cost iteration curves under five scenarios.

Specifically, the space-only scheduling scenario achieves a final cost of 1,076,638.32 CNY, corresponding to 0.51% reduction. The time-only scheduling scenario reduces the cost to 1,058,636.49 CNY, with a 2.18% reduction. With the integration of both temporal and spatial scheduling, the cost decreases further to 1,056,563.15 CNY, achieving a reduction of 2.37%. When energy storage is incorporated, the system cost is further reduced to 1,042,436.52 CNY, corresponding to a total reduction of 3.67%. A cross-scenario comparison reveals that space-only scheduling provides limited cost improvement because it merely redistributes workloads among data centers without altering their allocation across high- and low-price periods, showing limited ability to reduce high-cost electricity consumption. In contrast, time-only scheduling directly adjusts task execution timing, shifting workloads from high- to low-price periods, which significantly reduces electricity procurement costs and serves as the primary source of cost savings. When spatial scheduling is incorporated to a greater extent, additional cost reduction is achieved. This indicates that cross-data-center workload redistribution can further optimize electricity price matching within the same time period, allowing workloads to be allocated to data centers with more favorable electricity prices or greater capacity, thereby creating additional savings potential. With the integration of energy storage, the system cost decreases further to 1,042,436.52 CNY. In this case, the system not only avoids high-price periods through task scheduling but also leverages energy storage to charge during low-price periods and discharge during high-price periods. As a result, the reduction in electricity procurement cost is no longer solely dependent on workload migration, but is achieved through coordinated optimization of both computing and power resources.

The task time-slot distributions under different scheduling scenarios shown in Figure 7 show that different scheduling strategies exert markedly different impacts on the workload structure. Before optimization, the workload curves of the three data centers are relatively smooth, with DC1 mostly ranging from about 120 to 160 tasks, DC2 from about 110 to

150 tasks, and DC3 from about 95 to 130 tasks. This indicates that the original workload profiles are mainly determined by task arrivals and are not closely correlated with TOU electricity prices. After optimization, all scenarios exhibit more evident temporal fluctuations and inter-data-center differences, although the regulation patterns differ.

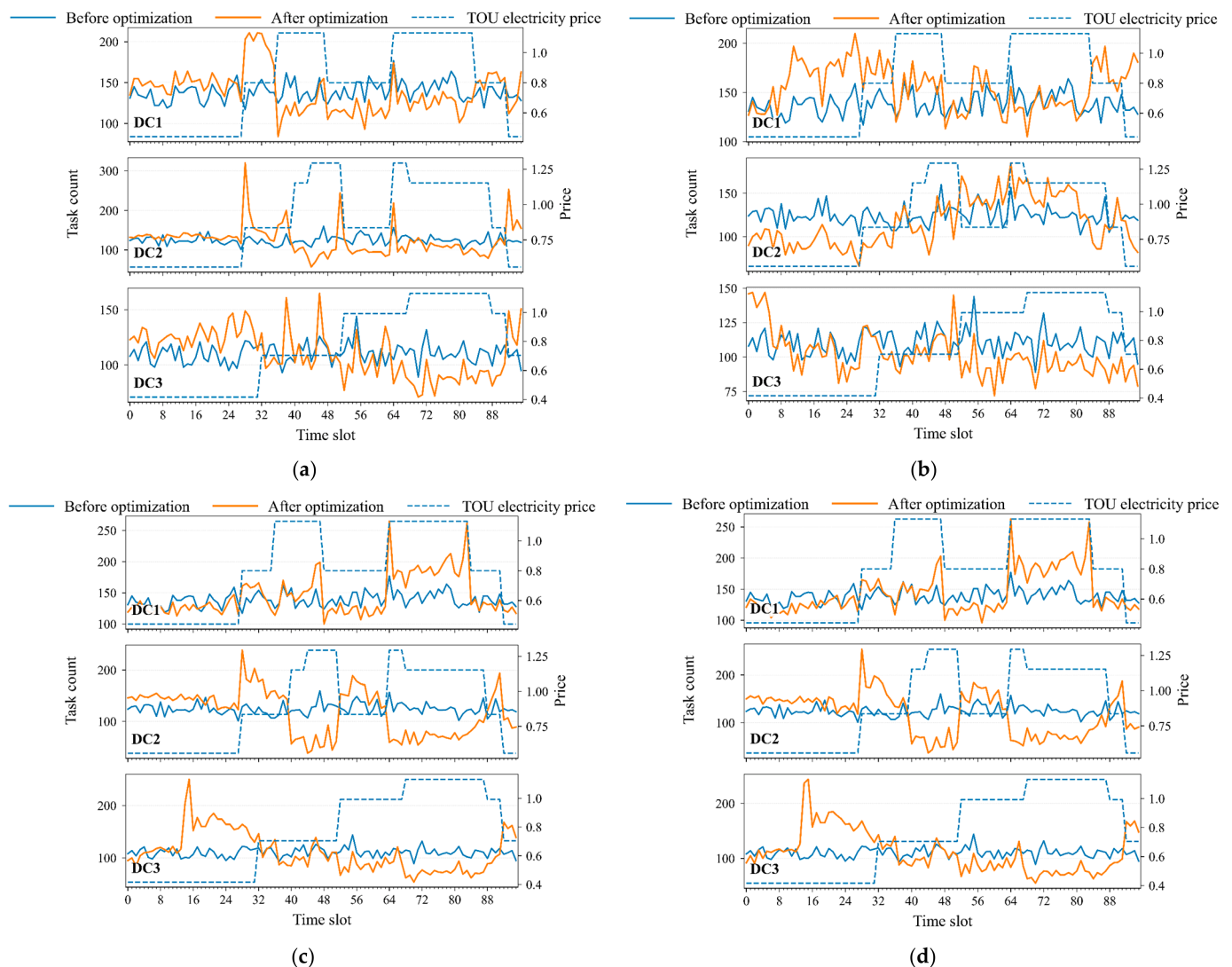


Figure 7. Task timeslot distribution under different scenarios. (a) time-only scheduling; (b) spatial-only scheduling; (c) spatiotemporal scheduling; (d) spatiotemporal scheduling with energy storage.

Under time-only scheduling, the adjustment is mainly reflected in intra-temporal reallocation within each data center. As shown in Figure 7a, DC1 shows obvious workload aggregation around time slots 28–34, with the peak exceeding 200 tasks, while the task count decreases to about 100–130 in the following high-price periods. DC2 also presents several local peaks around time slots 28, 48, and 90, with the highest peak approaching 300 tasks. For DC3, the task count in later high-price periods is mostly reduced to about 80–100. This indicates that temporal shifting can effectively respond to price signals by transferring part of the workload away from high-price periods, but it may also increase load concentration within specific low-price or feasible execution windows.

In contrast, space-only scheduling is characterized by inter-data-center redistribution. As shown in Figure 7b, DC1 undertakes more tasks after optimization, with several time slots reaching about 170–200 tasks. Meanwhile, DC3 decreases to about 80–100 tasks during time slots 55–90, which is generally lower than its pre-optimization level. DC2 also carries

more tasks in some middle and later periods, reaching about 140–170 tasks during time slots 60–80. Therefore, space-only scheduling mainly exploits regional electricity price differences and capacity differences among data centers. However, its compression effect on workloads during high-price periods remains limited, because this strategy does not alter the temporal distribution of tasks.

As spatiotemporal scheduling is adopted, the workload distribution exhibits a more pronounced structural reconfiguration. As shown in Figure 7c, the task count of DC2 is significantly reduced during time slots 40–50 and 66–86, with some slots decreasing to about 50–80 tasks. DC3 also remains mostly at about 60–100 tasks during time slots 55–88, while a clear task peak appears around time slot 15, exceeding 200 tasks. Meanwhile, DC1 undertakes more tasks during time slots 66–84, with some peaks approaching 250 tasks. This indicates that spatiotemporal scheduling jointly coordinates temporal shifting and spatial migration, rather than simply moving tasks within a single data center, thereby achieving a deeper level of workload restructuring.

After incorporating energy storage, the overall task-side distribution remains largely consistent with the spatiotemporal scheduling scenario without storage. As shown in Figure 7d, DC2 still maintains relatively low task levels during time slots 40–50 and 66–86, DC3 is mostly lower than the pre-optimization level in the later periods, and DC1 continues to undertake more shifted tasks. This suggests that energy storage does not directly change the task-count distribution significantly. Its main role lies on the power side, where low-price charging and high-price discharging reduce grid purchasing pressure and complement task scheduling.

Overall, Figure 7 shows that time-only scheduling mainly changes task execution time and may lead to local workload peaks; space-only scheduling mainly changes task execution location and enables inter-data-center workload redistribution; spatiotemporal scheduling produces a more evident workload reconstruction by combining both dimensions; and energy storage further improves system operation mainly through power-side regulation.

The power time-slot distributions in Figure 8 show the changes in power profiles under different scheduling scenarios. Compared with the task-count distributions in Figure 7, the power curves more directly reflect the impact of scheduling strategies on power-side operation. Before optimization, the power profiles of the three data centers are relatively smooth, with DC1 mostly maintained at about $2.20 \sim 2.35 \times 10^4$ kW, DC2 at about $1.76 \sim 1.83 \times 10^4$ kW, and DC3 at about $1.15 \sim 1.24 \times 10^4$ kW. This indicates that the original power demand is mainly determined by natural task arrivals and execution processes. After optimization, the correspondence between power profiles and TOU electricity prices becomes more evident, showing a segmented pattern with higher power in low-price periods and lower power in high-price periods.

Under time-only scheduling, the power variation mainly results from task shifting within each original data center. As shown in Figure 8a, DC1 approaches 2.40×10^4 kW during low-price periods, while it is reduced to about $2.05 \sim 2.10 \times 10^4$ kW during high-price periods such as time slots 36–48 and 64–88. DC2 is also reduced from about 1.8×10^4 kW to about $1.55 \sim 1.65 \times 10^4$ kW in high-price intervals, while DC3 remains around $1.08 \sim 1.12 \times 10^4$ kW in later high-price periods. This shows that time-only scheduling can effectively reduce power demand during high-price periods, but may cause step-like changes near price transition points.

Under space-only scheduling, power regulation is mainly reflected in workload redistribution among data centers. In Figure 8b, the optimized power of DC1 increases overall and mostly stays at about $2.33 \sim 2.40 \times 10^4$ kW. By contrast, DC2 mostly remains around $1.65 \sim 1.80 \times 10^4$ kW, and DC3 is also lower than the pre-optimization level in some periods, at about $1.13 \sim 1.20 \times 10^4$ kW. This indicates that space-only scheduling

transfers part of the workload to data centers with better economic or capacity conditions. However, since task execution time is not changed, its compression effect during high-price periods is less obvious than that of time-only scheduling.

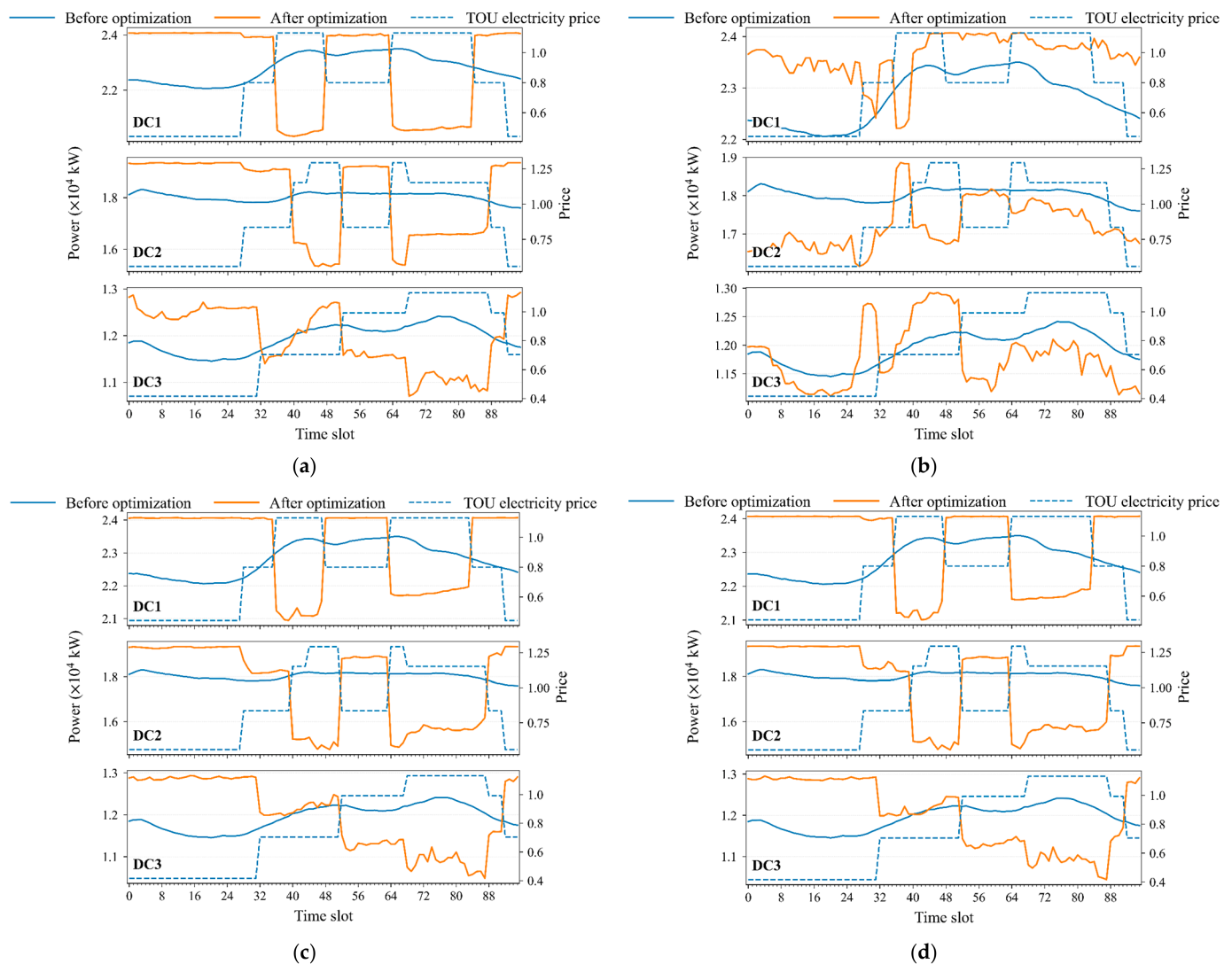


Figure 8. Power Timeslot Distribution Under Different Scenarios. (a) time-only scheduling; (b) spatial-only scheduling; (c) spatiotemporal scheduling; (d) spatiotemporal scheduling with energy storage.

Under spatiotemporal scheduling, the power curves show more evident peak-shaving and valley-filling characteristics. As shown in Figure 8c, DC1 maintains a high power level during low-price periods, close to 2.40×10^4 kW, while it stays around $2.10 \sim 2.20 \times 10^4$ kW during high-price periods. DC2 is reduced to about $1.50 \sim 1.58 \times 10^4$ kW during time slots 36–48 and 64–88, and DC3 remains around $1.06 \sim 1.14 \times 10^4$ kW in later high-price periods. Compared with single-dimensional scheduling, spatiotemporal scheduling jointly utilizes time-varying electricity prices and inter-data-center differences, leading to a more sufficient structural adjustment of power distribution.

After incorporating energy storage, the power distribution in Figure 8d is generally close to that in Figure 8c, indicating that energy storage does not significantly change the power structure formed by spatiotemporal scheduling. The data centers still maintain the basic feature of higher power during low-price periods and lower power during high-price periods. Compared with the scenario without energy storage, energy storage mainly makes

local power curves smoother and alleviates abrupt changes in some periods, serving more as an auxiliary regulation measure.

Overall, Figure 8 show that different scheduling strategies have distinct impacts on the power side. Time-only scheduling can directly reduce power during high-price periods but may cause step-like fluctuations; space-only scheduling mainly changes the power-sharing proportions among data centers; spatiotemporal scheduling produces more effective peak shaving and valley filling; and energy storage further complements task scheduling through power-side regulation.

3.4. Algorithm Effectiveness Analysis

To validate the effectiveness of the proposed method, NSGA-II, PSO, and the proposed HC-ALNS algorithm are applied to solve Scenario 5. The corresponding cost convergence curves are presented in Figure 9. In addition to the convergence curves, the wall-clock time required by each algorithm to reach the stopping iteration is also recorded to evaluate the computational overhead.

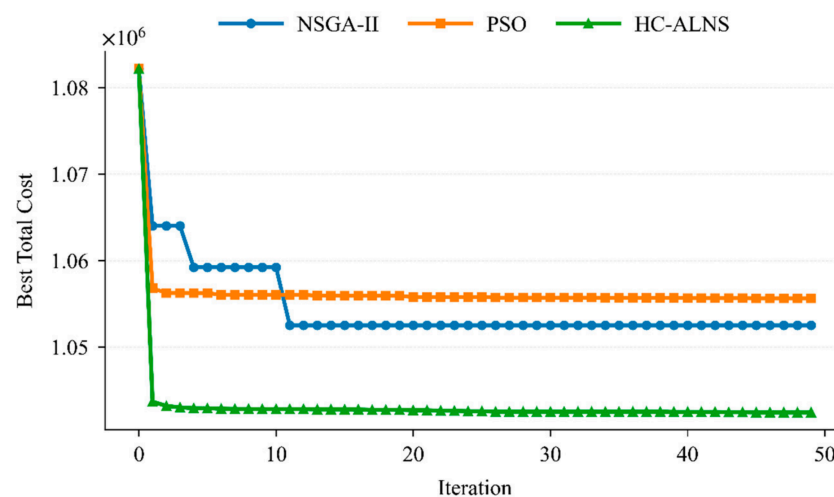


Figure 9. Cost iteration comparison of different algorithms.

As shown in the aforementioned figure, all three algorithms achieve a rapid cost reduction during the initial stage; however, their subsequent convergence behaviors differ significantly. The relevant iteration data are presented in Table 2. Specifically, PSO enters a plateau phase relatively early after an initial decrease, with limited further improvement, indicating a tendency toward premature convergence. Although PSO requires the shortest wall-clock time of 275.9 s, its final cost is 1,055,640 CNY, corresponding to a cost reduction of only 2.45%. In contrast, NSGA-II continues to exhibit gradual improvement in the mid-to-late stages, although its overall convergence speed remains relatively slow. This is also reflected in its wall-clock time of 1220.1 s, which is the longest among the three algorithms, while its final cost reduction is 2.74%. Compared with these methods, the HC-ALNS algorithm not only achieves a substantial cost reduction in the early stage but also maintains continuous improvement in solution quality during subsequent iterations. Although it reaches the optimal solution at a relatively later stage, this behavior suggests that the algorithm is less prone to being trapped in local optima and possesses stronger global exploration capability. HC-ALNS obtains the lowest final cost of 1,042,445 CNY with a cost reduction of 3.67%, and its wall-clock time is 762.3 s, which is lower than that of NSGA-II while achieving better solution quality.

Overall, HC-ALNS demonstrates superior performance in solution quality and overall computational cost-effectiveness compared to the benchmark algorithms, validating

its effectiveness and advantage in solving the multi-data-center coordinated scheduling problem. Although PSO has the shortest absolute running time, its optimization effect is limited. Therefore, HC-ALNS provides a better balance between computational overhead and cost reduction.

Table 2. Performance comparison of different algorithms.

Algorithm	Baseline Cost	Final Cost	Stop Iteration	Wall-Clock Time	Reduction (%)
HC-ALNS	1,082,175	1,042,445	27	762.3	3.67
NSGA-II	1,082,175	1,052,514	12	1220.1	2.74
PSO	1,082,175	1,055,640	21	275.9	2.45

As shown in Figure 10, the proposed algorithm demonstrates significant optimization capability in both task scheduling and power-side coordinated control. At the load level, the originally smooth load profiles of each data center are reshaped into distinctly segmented patterns post-optimization: loads are significantly reduced during high electricity price periods, increasing correspondingly during low- and mid-price periods. This indicates that the algorithm can globally reorganize flexible tasks in an orderly manner, achieving temporal load shifting. Meanwhile, the variation in purchased grid power is more pronounced than that of the load. It decreases more sharply during high-price periods and increases during low-price periods, reflecting the coordinated operation of energy storage charging/discharging and grid purchasing strategies. This enables the system to reduce grid dependence during high-price periods while proactively shifting energy usage to low-price periods. Correspondingly, the slot-wise cost distribution in Figure 11 validates the effectiveness of the scheduling strategy: costs are significantly reduced during high-price periods, while only slight increases are observed during low-price periods, resulting in an overall improvement in the cost structure. This demonstrates that the algorithm effectively mitigates the impact of high-price periods on total cost through task rescheduling and storage coordination.

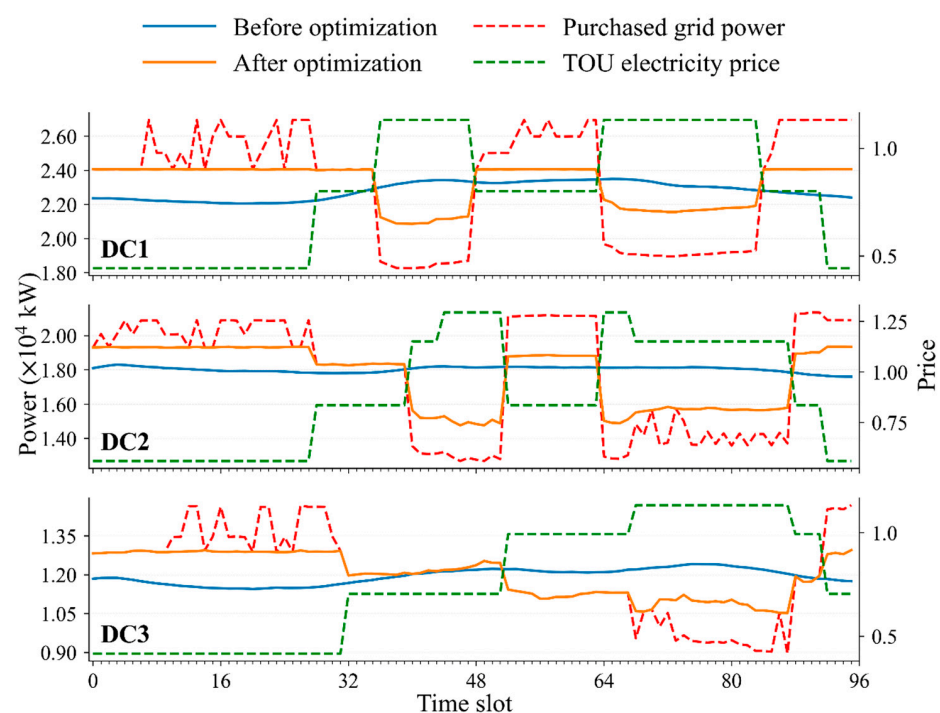


Figure 10. Purchased grid power comparison.

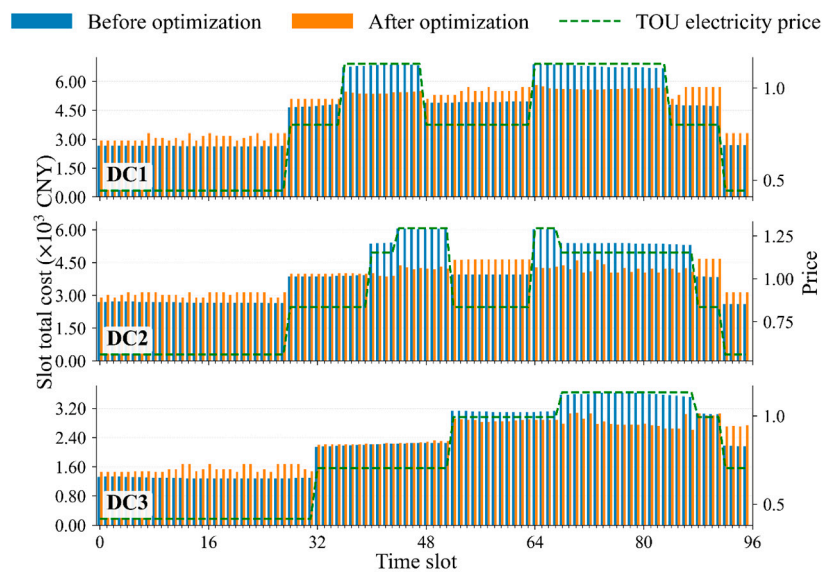


Figure 11. Total slot cost comparison for three data centers.

Overall, the algorithm establishes a coordinated optimization mechanism across task scheduling, energy storage control, and grid purchasing decisions, achieving load restructuring and temporal cost shifting, thereby validating its effectiveness in multi-data-center scheduling.

3.5. Sensitivity and Robustness Analysis

To further characterize the impact mechanisms of key parameters with regard to system performance, this subsection conducts a sensitivity analysis focusing on the proportion of transferable workloads and the degree of electricity price fluctuation. For data centers, transferable workloads serve as the primary carrier for peak shaving, valley filling, and cross-data-center coordinated scheduling, while electricity price variation provides a direct economic incentive for workload migration. Therefore, under the premise of a fixed total computational workload, the effects of these two factors on cost reduction are systematically analyzed by adjusting the transferable workload factor and the electricity price fluctuation factor.

Both parameters are varied within the range of $[-0.2, 0.2]$. As shown in Figure 12, the cost reduction generally exhibits a monotonic trend with respect to both factors, with the electricity price fluctuation factor exerting a more pronounced influence. As the degree of price fluctuation increases, the peak–valley price gap widens. This allows the scheduling strategy to more effectively exploit price signals and shift workloads toward low-price periods, thereby significantly enhancing the system’s cost-saving potential. Overall, the cost reduction shows an approximately linear increase with respect to the electricity price fluctuation factor, indicating that the pricing mechanism plays a dominant role in driving optimization benefits. In contrast, the impact of the transferable workload factor demonstrates a clear diminishing marginal effect. At relatively low levels of transferable workload, increasing its proportion enhances scheduling flexibility and yields noticeable cost reductions. However, as the proportion continues to grow, the improvement gradually saturates due to system capacity constraints and the inherent temporal–spatial distribution of workloads, reducing the sensitivity to further changes in this parameter. Moreover, when the proportion of transferable workload decreases, the system cost initially experiences a slight reduction and then enters a plateau phase. This suggests that the proposed model can adaptively adjust scheduling strategies under different workload compositions to maintain

near-optimal operation. Additionally, as the peak–valley price gap narrows, the influence of workload structure on system cost diminishes.

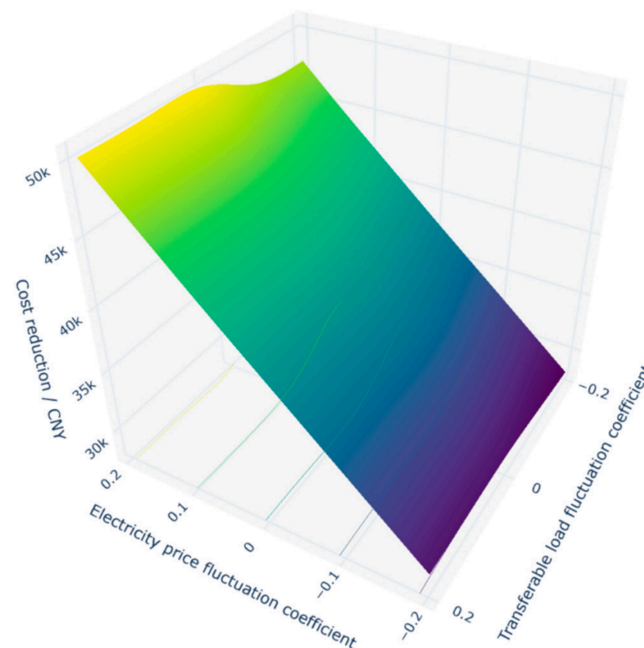


Figure 12. Sensitivity surface of cost reduction to key system parameters.

These results demonstrate that the proposed scheduling model maintains stable optimization performance under key parameter perturbations, exhibiting strong robustness and adaptability.

4. Discussion

This paper addresses the mathematical optimization problem arising from the coupled scheduling of computational workloads and power-side operation in multi-data-center systems. A multi-data-center mixed-integer nonlinear programming model (MDC-MINLP) is formulated for task-level computation–power coordinated scheduling, in which binary task scheduling decisions, device-level computing utilization, nonlinear power consumption characteristics, energy storage dynamics, grid procurement, and quality-of-service constraints are jointly represented. To solve the resulting large-scale discrete–continuous coupled problem, an Adaptive Large Neighborhood Search algorithm with Hierarchical Collaboration (HC-ALNS) is developed. The main findings are summarized as follows:

- By explicitly modeling task temporal flexibility, spatial migration capability, and flexible queue constraints, the proposed MDC-MINLP provides a fine-grained feasible region for workload restructuring. The optimization results show that task execution decisions are no longer distributed only according to original arrival patterns, but are reorganized according to electricity price signals and system constraints. Consequently, workloads are reduced during high-price periods and reassigned to low-price periods, indicating that the model can effectively transform task flexibility into cost-oriented scheduling decisions.
- The comparison among different scheduling scenarios verifies the contribution of each decision dimension to the objective value. Temporal scheduling provides the primary reduction in total operating cost by shifting flexible workloads away from high-price periods, achieving a cost reduction of 2.18%. On this basis, spatial scheduling further expands the feasible solution space by enabling cross-data-center workload redistribution, increasing the reduction to 2.37%. When energy storage operation is incorporated,

the model further couples workload allocation with charge–discharge decisions, using the “charge at low price and discharge at high price” mechanism to improve the power-side cost structure. As a result, the total cost reduction reaches 3.67%, demonstrating that the joint optimization of task scheduling and power operation yields a stronger objective improvement than single-dimensional scheduling.

- The proposed HC-ALNS algorithm shows better suitability for the structure of the MDC-MINLP than general-purpose metaheuristics. Through task action-space reconstruction, surrogate objective screening, accurate power-layer evaluation, and adaptive search intensity adjustment, HC-ALNS maintains continuous improvement in later iterations and reduces the probability of premature convergence. Compared with NSGA-II and PSO, HC-ALNS achieves additional cost reductions of 0.93% and 1.22%, respectively, indicating stronger convergence performance and higher final solution quality for the studied discrete–continuous coupled optimization problem.

Furthermore, the sensitivity analysis provides insight into the influence of key parameters on the optimal objective value. The electricity price fluctuation factor has a significant positive effect on cost reduction, indicating that the time-varying price coefficient is a dominant parameter driving the optimization benefit. In contrast, the proportion of transferable workloads exhibits diminishing marginal returns. When the amount of schedulable workload reaches a certain level, further increasing task flexibility only slightly enlarges the effective feasible region and contributes less to additional cost reduction. Under different parameter perturbations, the proposed MDC-MINLP and HC-ALNS maintain stable optimization performance, demonstrating robustness with respect to changes in price signals and workload flexibility.

These results indicate that the proposed mathematical modeling and solution framework can effectively describe and solve the coordinated operation problem of multiple data centers under time-of-use electricity pricing. By integrating task scheduling constraints, nonlinear power consumption relationships, energy storage dynamics, and operating cost components into a unified MDC-MINLP, the model captures the essential coupling between computation-side and power-side decisions. Meanwhile, HC-ALNS provides an efficient problem-oriented heuristic for obtaining high-quality feasible solutions in a large-scale mixed-integer nonlinear search space.

Future research may further extend the proposed model by incorporating renewable energy generation, uncertainty in workload arrivals and electricity prices, and real-time rolling optimization mechanisms. In addition, the scalability of the proposed method for larger networked data center systems, as well as its integration with electricity market participation and stochastic optimization frameworks, remains an important direction for further study.

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Conflicts of Interest: Author Liaoyi Ning was employed by the State Grid Liaoning Electric Power Supply Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Appendix A

Table A1. Heterogeneous Parameters of Data Centers.

Parameter	Symbol	DC1	DC2	DC3	Unit
Computing capacity	C_k	20,000	15,000	9000	CPU
Flexible queue capacity	$\bar{Q}_k$	4300	3000	2000	CPU
Aggregated idle IT power	p_k^{idle}	13,000	10,200	6700	kW
Aggregated full-load IT power	p_k^{full}	18,200	14,950	10,200	kW
Nonlinear IT power coefficient	τ_i	1.2	1.2	1.2	-
Base cooling power	$p_k^{cool,0}$	1200	950	650	kW
Cooling coefficient	α_k^{cool}	0.24	0.22	0.2	-
Base distribution loss	$p_k^{loss,0}$	380	300	210	kW
Distribution loss coefficient	α_k^{loss}	0.03	0.03	0.03	-
Grid power limit	$p_k^{grid,max}$	29,000	21,500	15,000	kW
Battery capacity	E_k^{bat}	20,000	16,000	12,000	kWh
Charging efficiency	η_k^{ch}	0.95	0.95	0.95	-
Discharging efficiency	η_k^{dis}	0.95	0.95	0.95	-
Minimum state of charge	SOC_k^{min}	0.1	0.1	0.1	-
Maximum state of charge	SOC_k^{max}	0.9	0.9	0.9	-
Initial state of charge	$SOC_{k,0}$	0.5	0.5	0.5	-
Max charging power	$p_k^{ch,max}$	3000	2500	1800	kW
Max discharging power	$p_k^{dis,max}$	3000	2500	1800	kW
Optimal utilization level	ρ^{opt}	0.7	0.7	0.7	-
Utilization deviation coefficient	γ	0.2	0.18	0.16	-
IT operation loss coefficient	λ_{IT}	0.06	0.06	0.06	CNY/(CPU·h)
Battery degradation coefficient	λ_{bat}	0.06	0.06	0.06	CNY/kWh
Migration cost coefficient	λ_{mig}	0.002	0.002	0.002	CNY/CPU
Number of equivalent cabinets	$\mathcal{I}_k$	1000	750	520	count
Idle IT power per cabinet	p_i^{idle}	13	13.6	12.88	kW
Full-load IT power per cabinet	p_i^{full}	18.2	19.93	19.62	kW

Table A2. Global Parameters and Algorithm Parameters.

Parameter	Symbol	Value	Unit
Number of data centers	$\mathcal{N}$	3	count
Scheduling horizon	$\mathcal{T}$	96	slot
Time-slot duration	Δt	0.25	h
Total number of task inputs	$\mathcal{M}$	30,637	task
Maximum waiting delay of time-shiftable type-1 tasks	$\bar{W}_m$	5	slot
Maximum waiting delay of time-shiftable type-2 tasks	$\bar{W}_m$	10	slot
Maximum waiting delay of time-shiftable type-3 tasks	$\bar{W}_m$	15	slot
Surrogate peak penalty coefficient	α_1	2600	-
Surrogate hard-peak penalty coefficient	α_2	380,000	-
Surrogate smoothing coefficient	α_3	24	-
Surrogate utilization-deviation coefficient	α_4	45	-

Table A2. Cont.

Parameter	Symbol	Value	Unit
Surrogate battery-friendly coefficient	α_5	0.14	-
Surrogate capacity-violation coefficient	α_6	50,000,000	-
Utilization warning threshold	ρ^{warn}	0.82	-
Hard utilization upper bound	ρ^{max}	0.9	-
Time-slot shift penalty coefficient	λ_{shift}	0.008	-
Flexibility weight for temporal shiftability	β_1	0.45	-
Flexibility weight for spatial shiftability	β_2	0.3	-
Congestion weight for average utilization	β_3	1.6	-
Congestion weight for peak utilization	β_4	2.2	-
Price sensitivity weight	β_5	2.8	-
Uphill tolerance ratio	ε_r^{up}	0.00015	-
Adaptive progress coefficient	μ_1	0.75	-
Adaptive stagnation coefficient	μ_2	0.18	-
Adaptive low-improvement coefficient	μ_3	0.42	-
Minimum effective improvement threshold	η_{min}	0.00012	-
Maximum number of iterations	R_{max}	48	iteration
Minimum improvement threshold	ε	0.000001	-

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